TOBB ETU

Economy & Technology University

BIL 495 / YAP 495

Project Management Plan (PMP)

Reference: IEEE 1058 / ISO/IEC 12207

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1. Project Overview

In Neonatal Intensive Care Units (NICUs); medical devices, sensors, and Hospital Information Systems (HIS) often operate independently and use incompatible data formats, which prevents clinicians from obtaining a unified, real-time view of newborns' medical conditions. Moreover, raw data from sensors are frequently noisy, incomplete, or non-standardized; which reduces the accuracy and reliability of AI-based diagnostic and monitoring systems.

Clinicians also lack an integrated platform that consolidates historical patient records, AI-generated diagnostic results, and updated clinical guidelines. As a result, clinical decision-making becomes slower, less consistent, and more prone to human error.

To address these issues, there is a clear need for a secure, interoperable, and real-time digital decision support infrastructure that can unify all NICU data sources—IoMT devices, HIS platforms, and AI diagnostic modules—under standardized protocols such as HL7, DICOM, and FHIR.

2. Organization and Roles

Team members:

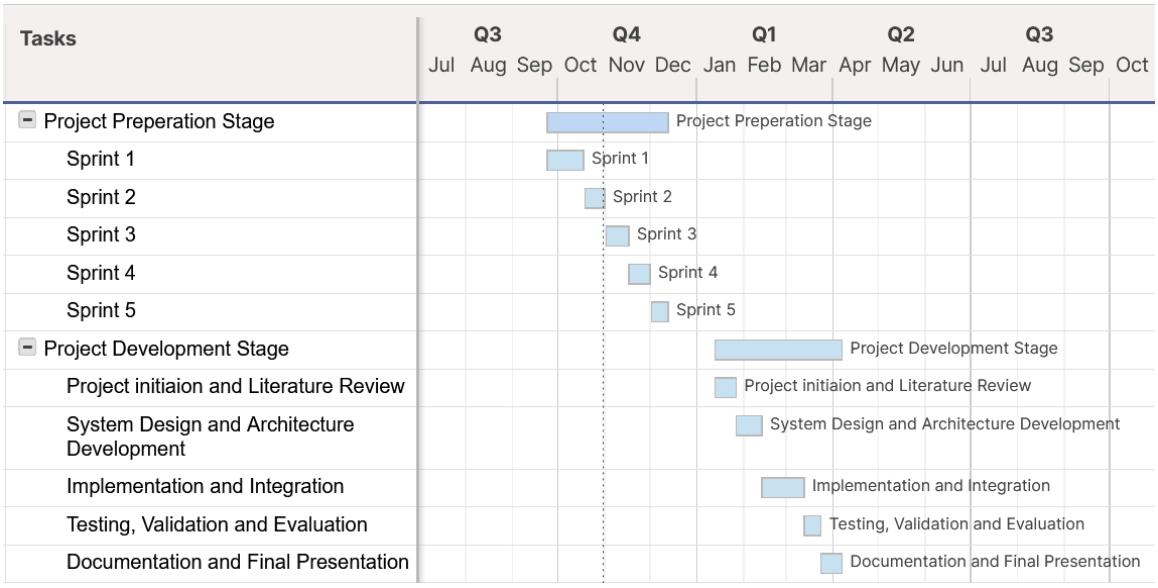
- Ali Belikırık
- Beyza Değirmenci
- İnan Özşahin
- Özge Doğan

3. Project Schedule

Project	Start Date	End Date	Duration
Project Preparation Stage	**2025-09-23**	**2025-12-12**	**80**
Sprint 1	2025-09-23	2025-10-17	24
Sprint 2	2025-10-18	2025-10-31	13
Sprint 3	2025-11-01	2025-11-16	15
Sprint 4	2025-11-16	2025-11-30	14
Sprint 5	2025-12-01	2025-12-12	11
Project Development Stage	**2026-01-12**	**2026-04-06**	**84**
Project Initiation and Literature Review	2026-01-12	2026-01-26	14

System Design and Architecture Development	2026-01-26	2026-02-12	17
Implementation and Integration	2026-02-12	2026-03-12	28
Testing, Validation and Evaluation	2026-03-12	2026-03-23	11
Documentation and Final Presentation	2026-03-23	2026-04-06	14

Gantt Chart



4. Budget and Resources

Resources

- **Software:** TBD
- **Hardware:** CPU based on x86 architecture, camera(s) for visual data, 32GB+ RAM, 256 GB+ Storage

- **Frameworks/Protocols:** HL7, FHIR, DICOM, RESTful APIs

- **AI Tools:** TensorFlow, PyTorch, Scikit-learn
- **Version Control:** GitHub
- **Collaboration Tools:** Trello, Microsoft Teams

5. Risk Management

The patient data may be at risk of unauthorized access, data breach and loss of patient data due to system vulnerabilities.

To mitigate such risks the data will be processed and protected to fit with security and encryption measures already in place. Strict user authentication will be applied.

6. Quality Assurance

6.1 Societal Benefits

The proposed secure, interoperable, and real-time digital decision support system will considerably enhance neonatal care outcomes by automating the data preprocessing for use in Hospital Information Systems (HIS) and AI-based diagnostic and monitoring systems. This will allow the clinicians to access required data on infant's health faster and let AI-based diagnostics provide better results by eliminating the data before it is fed to the models which in turn will improve the care that can be provided. In the long term this may lead to improved survival rates for infants.

6.2 Economic Constraints

The project will prioritize open-source software and scalable cloud infrastructure to minimize costs. The proposed AI model shall be capable of performing inference on standard CPU hardware without requiring GPU acceleration.

6.3 Legal and Ethical Compliance

The system will strictly adhere to standards on medical data management and patient care. Patient confidentiality and informed consent will be priority and each part of the system will be built in accordance with these ethical principles.

7. Appendix

8. References

1. *Health Level Seven International (HL7) Standards, www.hl7.org*
2. *FHIR (Fast Healthcare Interoperability Resources), https://www.hl7.org/fhir/*
3. *DICOM Standards Committee, Digital Imaging and Communications in Medicine (DICOM)*